

BHTTP/1

Binary HTTP · version 1 · compact interoperability specification

Purpose. BHTTP/1 carries HTTP-style file requests and responses as binary frames over one long-lived TCP connection. All integers are unsigned, big-endian; all lengths count bytes. MUST / SHOULD / MAY have their usual RFC 2119 meaning.

1 Connection and handshake

Transport: TCP, default port 9000. The client sends eight preface bytes: 42 48 54 50 01 00 0D 0A (“BHTTP”, major 1, minor 0, CRLF). A valid server echoes the preface before any frame. Bad magic/CRLF causes GOAWAY(400); unsupported major causes GOAWAY(505); both close the connection. Unknown minor versions are accepted. The client SHOULD send GOAWAY(0) before closing. After 30 seconds without a frame, the server MAY send GOAWAY(408).

2 Fixed frame header

Every frame begins with this eight-byte header: the payload starts immediately after it.

Length (24 bits) Type (8) Flags (8) Request ID (24 bits) Payload (Length bytes)		
Field	Width	Meaning
Length	24 bits	0–16,777,215 payload bytes; larger bodies are split.
Type	8 bits	256 possible types; unknown types are extensions.
Flags	8 bits	END_STREAM = 0x01; REUSE_HEADERS = 0x02; other bits ignored.
Request ID	24 bits	Identifies pipelined/interleaved requests. 0 is connection-level and used only by GOAWAY.

Frame types

Type	Name	Direction	Rules
0x01	REQUEST	Client → server	ID ≥ 1, strictly increasing; END_STREAM / REUSE_HEADERS.
0x02	RESPONSE	Server → client	Answers matching request ID; may carry END_STREAM.
0x03	DATA	Both	Raw body bytes; final DATA carries END_STREAM.
0x07	GOAWAY	Both	ID 0; last ID (3), code (2), optional UTF-8 reason.
other	Unknown	Either	MUST read and discard Length bytes, then continue.

3 Request and header encoding

REQUEST payload: method (1) | path_len (2) | path (UTF-8) | header block. The path MUST begin with “/”; query text after “?” is ignored; a directory path serves index.html. Method codes: 1 GET, 2 HEAD, 3 POST, 4 PUT, 5 DELETE, 6 OPTIONS. File servers support GET and HEAD.

Header block	Encoding
Count	1 byte; present even when zero.
Static name	index 1–10, then value_len (2) and value bytes.
Literal name	tag 0 name_len (1) lowercase ASCII name value_len (2) value.
Static table	1 host · 2 user-agent · 3 accept · 4 if-none-match · 5 content-type · 6 content-length · 7 last-modified · 8 etag · 9 cache-control · 10 date

REUSE_HEADERS: flag 0x02 starts with the previous request’s effective headers; this frame’s entries replace/add values. After a malformed request the saved set is cleared.

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Responses, lifecycle, errors, and a wire example

4 Response and body framing

RESPONSE payload: status (2) | header block. Status is the HTTP number itself (200, 304, 400, 404, 405, 505, ...). The body is carried in DATA frames and concatenated byte-for-byte; no escaping or text conversion is performed, so binary files are safe. HEAD, 304, and empty bodies set END_STREAM on RESPONSE. Otherwise the last DATA frame sets END_STREAM. DATA frames SHOULD be 16 KiB or less.

5 Pipelining and ordering

The client MAY send several REQUEST frames without waiting. The server MAY respond in any order and MAY interleave frames from different responses. The client matches frames by Request ID, never by arrival order. Paths that escape the server root return 404.

6 Error and closing rules

Condition	Result
Malformed request payload: short field, invalid UTF-8/path, reserved index, trailing bytes	RESPONSE 400 for that ID; connection stays open.
REQUEST payload over 65,535 bytes	Read and discard it, then RESPONSE 400.
Missing file or path escapes root	RESPONSE 404.
Unsupported method	RESPONSE 405.
Bad preface / unsupported major / invalid request ID	GOAWAY 400 / 505 / 400, then close.
EOF in the middle of a frame	Close connection.

Client exit status: 0 when all responses are below 400; 4 for any 4xx; 5 for any 5xx (5 wins); 2 for a connection/protocol error; 1 for bad usage. Bodies are still delivered on 4xx/5xx.

7 Worked example · GET /hello.txt

The sample file is 19 bytes: Hello, I am Aryen.\n.

Direction / bytes	Interpretation
C→S 42 48 54 50 01 00 0D 0A	BHTTP/1 preface.
C→S 00 00 31 01 01 00 00 01	Length 49, REQUEST, END_STREAM, ID 1. Payload is GET, path length 10, /hello.txt, and host/user-agent/accept headers.
S→C 42 48 54 50 01 00 0D 0A	Server preface echo.
S→C 00 00 8B 02 00 00 00 01 00 C8 ...	Length 139, RESPONSE, ID 1, status 200; headers include content-length=19.
S→C 00 00 13 03 01 00 00 01 48 65 ... 0A	Length 19, DATA, END_STREAM, ID 1; raw file bytes.
C→S 00 00 05 07 00 00 00 00 00 00 01 00 00	GOAWAY(0): last request ID 1, graceful close.

Interoperability rule: an implementation is correct only when an independently written client and server communicate using this document alone.